

DynamoDB Training Program

Module 1: Introduction to DynamoDB

- ☐ Overview of AWS DynamoDB
 - ☐ Features of DynamoDB
 - ☐ Use Cases for DynamoDB
 - ☐ Differences Between SQL and NoSQL Databases
 - ☐ Setting Up AWS Account and DynamoDB Console
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Module 2: Basics of DynamoDB

- ☐ Understanding Tables, Items, and Attributes
 - ☐ Creating and Managing Tables
 - ☐ Primary Key: Partition Key and Sort Key
 - ☐ Data Types in DynamoDB (String, Number, Boolean, etc.)
 - ☐ DynamoDB Limits and Pricing Overview
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Module 3: Advanced Table Design

- ☐ Secondary Indexes (Global and Local)
- ☐ Designing Scalable Table Structures

- ☐ Optimizing Read and Write Operations
 - ☐ Query vs Scan: Best Practices
 - ☐ Partitioning and Data Distribution
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Module 4: Data Operations in DynamoDB

- ☐ CRUD Operations: Create, Read, Update, Delete
 - ☐ Batch Write and Batch Read
 - ☐ Working with AWS SDKs (Python, Node.js, or Java)
 - ☐ Using the DynamoDB CLI
 - ☐ Pagination and Filtering Data
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Module 5: DynamoDB Streams and Transactions

- ☐ Introduction to DynamoDB Streams
 - ☐ Working with Streams to Track Changes
 - ☐ DynamoDB Transactions: ACID Compliance
 - ☐ Use Cases for Transactions in DynamoDB
 - ☐ Integrating Streams with AWS Lambda
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Module 6: Security and Access Control

- ☐ AWS IAM Roles and Policies for DynamoDB
 - ☐ Securing DynamoDB with Encryption at Rest and in Transit
 - ☐ Fine-Grained Access Control
 - ☐ Monitoring and Logging with AWS CloudWatch
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Module 7: Backup, Restore, and Global Tables

- ☐ Setting Up On-Demand Backups and Point-in-Time Recovery
 - ☐ Restoring Data from Backups
 - ☐ Introduction to Global Tables
 - ☐ Multi-Region Replication and Use Cases
 - ☐ Managing Global Tables
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Module 8: Performance Optimization

- ☐ Best Practices for Read and Write Capacity Units (RCU/WCU)
 - ☐ On-Demand vs Provisioned Capacity Mode
 - ☐ Caching with DynamoDB Accelerator (DAX)
 - ☐ Handling Large Data Sets and Hot Partitions
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Module 9: Integrations and Real-World Applications

- ☐ Integrating DynamoDB with Other AWS Services (S3, Lambda, API Gateway)
 - ☐ Building Serverless Applications with DynamoDB
 - ☐ Implementing Real-World Applications (e.g., Chat App, E-Commerce Backend)
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Module 10: Capstone Project

- ☐ Designing and Building a Scalable Application Using DynamoDB
- ☐ Implementing Data Models, Queries, and Indexing
- ☐ Using DynamoDB Streams and Lambda for Real-Time Updates
- ☐ Deploying the Application in a Serverless Architecture